Input: [A], [B], [C], [D], [E], [f], [G], [H]

1. NAIVE

- one thread processes everything

A
A+B
A+B+C

[A], [B], [C], [D], [E], [f], [G], [H]

↓

Thread 0

↓

output

threads: 1
shmem: no
syncthreads: no
parallel reduction: no

2. Interleaved shmem

index: 0 1 2 3 4 5 6 7

value: [A], [B], [C], [D], [E], [f], [G], [H]

R1 → stride = 1

T0: 0 and 1) T0 T2: 4 and 5) T4

T1: 2 and 3) T2 T3: 6 and 7) T6

R2 → stride = 2

T0: 0 and 2) T0

T4: 4 and 6) T4

(active threads on each round)

R3 → stride = 4

T0: 0 and 4 ⇒ [A], [B], [C], [D], [E], [f], [G], [H]

T0

3. Sequential address

index : 0 1 2 3 4 5 6 7

value : [A], [B], [C], [D], [E], [F], [G], [H]

R1 -> stride=4

T0 : A and E) T0

T1 : B and F) T1

T2 : C and G) T2

T3 : D and H) T3

R2 -> stride = 2

T0 : 0 and 2) T0

T1 : 1 and 3) T1

R3 -> stride = 1

T0 : 0 and 1) T0

[A], [B], [C], [D], [E], [F], [G], [H]

T0

! Interleaved : T0 -- T2 -- T4 -- T6 --

Sequential : T0 T1 T2 T3 -- -- -- --

4. warp-shuffle

-> mathematical tree looks same as sequential

the difference is where the values are stored

shmem : thread register

↓

shared[tid]

warp : thread register

↓

another lane reads that register directly

another thread reads shared[]

⇒

```
Naive
  ↓ introduce parallel threads
Interleaved shared memory
  ↓ organize active threads better
Sequential addressing
  ↓ avoid most shared-memory communication
Warp shuffle
```

the Term AtomicAdd();

before `out[blockIdx.x] = shared[0];`

↳ out [256, 256, 256, 232]

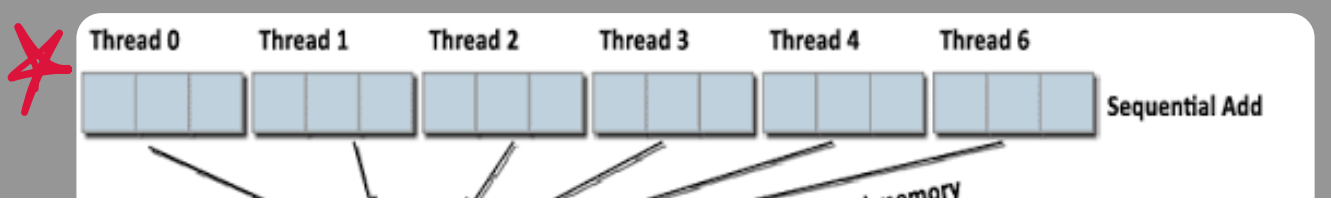
↳ this requires another reduction second kernel launch!

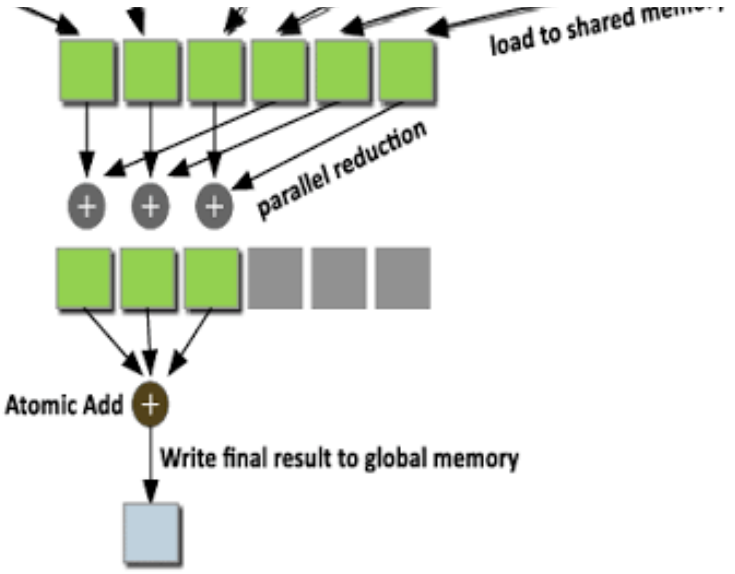
with atomic add `atomicAdd(out, shared[0])`

↳ out starts at 0

+ 256
+ 256
+ 256
+ 232 = 1000

- Shared-memory reduction still combines threads **inside each block**.
- `atomicAdd` combines results **between blocks**.
- Output changes from an array of partial sums to one final scalar.
- `cudaMemset` is necessary because atomic addition must start from zero.
- The tradeoff is that atomic updates to the same address happen one at a time, so many blocks can create contention. For this simple task, one atomic per block is reasonable.




load to shared mem
parallel reduction
Atomic Add
Write final result to global memory